Evaluation runs: calc (calculator-disel)

generated 2026-11-05 from brain-data cfc25f174 2026-11-05 13:40:01 +0100

Every campaign run on this benchmark, one row per instance (variant, specification, communication model, size) and engine; the newest run of an instance wins. Expected verdicts (the rule file): no entry in the rule file (the README states the expectation). Blue rows are at the current pins (TraceForge 367cdc6, whose predecessor 9a0629e is equivalent for every engine but the join; merge join 367cdc6; JMC 3c43b2d2): a rerun gives the same result. Red rows are superseded (an older pin, or the in-memory join that the merge join replaces) and their rerun is pending. A planned instance without a measurement is not a row: it is listed under "Not run". G_{spec}, G_{impl} = the graphs of the specification and of the implementation (the stateless engine's own count of specification graphs explored, repeats included, is G^{expl}_{spec}); G_x = the implementation graphs no specification graph covers (violating pairs); T = wall seconds; M = peak resident MiB; tier = the run's time and memory limits; * marks a verdict other than the expected one; 🕒 = the time tier was reached; oom = the memory limit was reached.

stateless

No run of the stateless engine on this benchmark.

stateful

No run of the stateful engine on this benchmark.

offline

offline: pair **batched**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2	6	4	conforms	0.0	4	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain05, 2026-11-02
3	20	8	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	70	16	conforms	0.3	4	10 min, 16 GB	0.3	0.1	0.0	in-memory	brain05, 2026-11-02
5	252	32	conforms	0.2	4	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
6	924	64	conforms	0.2	10	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
7	3432	128	conforms	0.7	34	10 min, 16 GB	0.5	0.1	0.2	in-memory	brain05, 2026-11-02
8	12870	256	conforms	1.7	138	10 min, 16 GB	0.8	0.1	0.9	in-memory	brain05, 2026-11-02
9	48620	512	conforms	5.3	566	10 min, 16 GB	1.7	0.1	3.6	in-memory	brain05, 2026-11-02
10	184756	1024	conforms	21.1	2328	10 min, 16 GB	6.0	0.2	15.1	in-memory	brain05, 2026-11-02

offline: pair **delegating**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
2	6	6	conforms	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
3	20	20	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	70	70	conforms	0.3	4	10 min, 16 GB	0.3	0.1	0.0	in-memory	brain05, 2026-11-02
5	252	252	conforms	0.2	4	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
6	924	924	conforms	0.3	14	10 min, 16 GB	0.1	0.2	0.1	in-memory	brain05, 2026-11-02
7	3432	3432	conforms	0.9	59	10 min, 16 GB	0.5	0.5	0.4	in-memory	brain05, 2026-11-02
8	12870	12870	conforms	2.8	243	10 min, 16 GB	0.8	1.1	1.7	in-memory	brain05, 2026-11-02
9	48620	48620	conforms	9.6	1008	10 min, 16 GB	1.7	2.4	7.2	in-memory	brain05, 2026-11-02
10	184756	184756	conforms	37.8	4133	10 min, 16 GB	6.0	8.2	29.5	in-memory	brain05, 2026-11-02

offline: pair **memo**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2	6	6	conforms	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
3	20	20	conforms	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	70	70	conforms	0.3	4	10 min, 16 GB	0.3	0.1	0.0	in-memory	brain05, 2026-11-02
5	252	252	conforms	0.2	4	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
6	924	924	conforms	0.3	16	10 min, 16 GB	0.1	0.2	0.1	in-memory	brain05, 2026-11-02
7	3432	3432	conforms	0.9	60	10 min, 16 GB	0.5	0.5	0.4	in-memory	brain05, 2026-11-02
8	12870	12870	conforms	2.5	243	10 min, 16 GB	0.8	0.7	1.7	in-memory	brain05, 2026-11-02
9	48620	48620	conforms	8.9	1008	10 min, 16 GB	1.7	1.7	7.2	in-memory	brain05, 2026-11-02
10	184756	184756	conforms	32.8	4132	10 min, 16 GB	6.0	5.6	26.8	in-memory	brain05, 2026-11-02

offline: pair **batched-swap**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	1	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2	6	4	1	violation	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
3	20	8	1	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	70	16	1	violation	0.3	4	10 min, 16 GB	0.3	0.1	0.0	in-memory	brain05, 2026-11-02
5	252	32	1	violation	0.2	4	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
6	924	64	1	violation	0.2	10	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
7	3432	128	1	violation	0.7	32	10 min, 16 GB	0.5	0.1	0.2	in-memory	brain05, 2026-11-02
8	12870	256	1	violation	1.7	138	10 min, 16 GB	0.8	0.1	0.9	in-memory	brain05, 2026-11-02
9	48620	512	1	violation	5.3	568	10 min, 16 GB	1.7	0.1	3.6	in-memory	brain05, 2026-11-02
10	184756	1024	1	violation	20.8	2327	10 min, 16 GB	6.0	0.2	14.7	in-memory	brain05, 2026-11-02

offline: pair **delegating-prev**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	2	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
2	6	6	6	violation	0.0	4	10 min, 16 GB	0.0	0.0	0.0	in-memory	brain05, 2026-11-02
3	20	20	20	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	70	70	70	violation	0.3	4	10 min, 16 GB	0.3	0.1	0.0	in-memory	brain05, 2026-11-02
5	252	252	252	violation	0.2	4	10 min, 16 GB	0.2	0.1	0.0	in-memory	brain05, 2026-11-02
6	924	924	924	violation	0.3	16	10 min, 16 GB	0.1	0.2	0.1	in-memory	brain05, 2026-11-02
7	3432	3432	3432	violation	0.9	62	10 min, 16 GB	0.5	0.5	0.4	in-memory	brain05, 2026-11-02
8	12870	12870	12870	violation	2.9	260	10 min, 16 GB	0.8	1.1	1.7	in-memory	brain05, 2026-11-02
9	48620	48620	48620	violation	9.7	1007	10 min, 16 GB	1.7	2.5	7.2	in-memory	brain05, 2026-11-02
10	184756	184756		violation	38.3	4129	10 min, 16 GB	6.0	8.5	29.8	in-memory	brain05, 2026-11-02

offline: pair **memo-ignore-args**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	2	violation	0.1	4	10 min, 16 GB	0.1	0.0	0.0	in-memory	brain05, 2026-11-02
2	6	6	6	violation	0.1	4	10 min, 16 GB	0.0	0.1	0.0	in-memory	brain05, 2026-11-02
3	20	20	20	violation	0.1	4	10 min, 16 GB	0.1	0.1	0.0	in-memory	brain05, 2026-11-02
4	70	70	70	violation	0.3	4	10 min, 16 GB	0.3	0.2	0.0	in-memory	brain05, 2026-11-02
5	252	252	252	violation	0.2	4	10 min, 16 GB	0.2	0.2	0.0	in-memory	brain05, 2026-11-02
6	924	924	924	violation	0.5	16	10 min, 16 GB	0.1	0.4	0.1	in-memory	brain05, 2026-11-02
7	3432	3432	3432	violation	0.9	63	10 min, 16 GB	0.5	0.4	0.3	in-memory	brain05, 2026-11-02
8	12870	12870	12870	violation	2.4	257	10 min, 16 GB	0.8	0.9	1.5	in-memory	brain05, 2026-11-02
9	48620	48620	48620	violation	8.5	1006	10 min, 16 GB	1.7	2.0	6.6	in-memory	brain05, 2026-11-02
10	184756	184756		violation	33.1	4135	10 min, 16 GB	6.0	5.7	27.1	in-memory	brain05, 2026-11-02

Not run (offline). planned but never scheduled (grid points beyond the ladder): 20 delegating bag, 20 batched-swap bag, 20 batched bag, 20 delegating-prev bag, 20 memo bag, 20 memo-ignore-args bag.

cross offline, Java implementation against the Rust specification

cross offline, Java implementation against the Rust specification: pair **batched**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
4	70	16	conforms	3.6	166	10 min, 16 GB	0.3	3.6	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **delegating**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
4	70	70	conforms	4.5	190	10 min, 16 GB	0.3	4.5	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **memo**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3	20	20	conforms	3.6	164	10 min, 16 GB	0.1	3.6	0.0	in-memory	brain01, 2026-11-02
4	70	70	conforms	3.8	158	10 min, 16 GB	0.3	3.8	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **batched-swap**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
4	70	16	1	violation	3.6	164	10 min, 16 GB	0.3	3.6	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **delegating-prev**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
3	20	20	20	violation	3.8	170	10 min, 16 GB	0.1	3.8	0.0	in-memory	brain01, 2026-11-02
4	70	70	70	violation	4.7	231	10 min, 16 GB	0.3	4.7	0.0	in-memory	brain01, 2026-11-02

cross offline, Java implementation against the Rust specification: pair **memo-ignore-args**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
3	20	20	20	violation	3.6	168	10 min, 16 GB	0.1	3.6	0.0	in-memory	brain01, 2026-11-02
4	70	70	70	violation	3.8	166	10 min, 16 GB	0.3	3.8	0.0	in-memory	brain01, 2026-11-02

Not run (cross offline, Java implementation against the Rust specification). refused (an input missing at run time): 1 bag batched, 1 bag delegating, 1 bag memo, 1 bag memo-ignore-args, 1 bag batched-swap, 1 bag delegating-prev, 2 bag batched, 2 bag memo, 2 bag delegating-prev, 2 bag memo-ignore-args, 2 bag delegating, 2 bag batched-swap, 3 bag delegating, 3 bag batched-swap, 3 bag batched.

cross offline, Rust implementation against the Java specification

cross offline, Rust implementation against the Java specification: pair **batched**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.1	156	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
2	6	4	conforms	3.4	158	10 min, 16 GB	3.4	0.0	0.0	in-memory	brain01, 2026-11-02
3	20	8	conforms	3.6	158	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain01, 2026-11-02
4	70	16	conforms	3.8	172	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair delegating, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.1	156	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-02
2	6	6	conforms	3.4	158	10 min, 16 GB	3.4	0.1	0.0	in-memory	brain01, 2026-11-02
3	20	20	conforms	3.6	158	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain01, 2026-11-02
4	70	70	conforms	3.8	172	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair memo, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.1	156	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
2	6	6	conforms	3.4	158	10 min, 16 GB	3.4	0.1	0.0	in-memory	brain01, 2026-11-02
3	20	20	conforms	3.6	158	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain01, 2026-11-02
4	70	70	conforms	3.8	172	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair batched-swap, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	1	violation	3.1	156	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
2	6	4	1	violation	3.4	158	10 min, 16 GB	3.4	0.1	0.0	in-memory	brain01, 2026-11-02
3	20	8	1	violation	3.6	158	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain01, 2026-11-02
4	70	16	1	violation	3.8	172	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair delegating-prev, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	2	violation	3.1	156	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
2	6	6	6	violation	3.4	158	10 min, 16 GB	3.4	0.0	0.0	in-memory	brain01, 2026-11-02
3	20	20	20	violation	3.6	158	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain01, 2026-11-02
4	70	70	70	violation	3.8	172	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain01, 2026-11-02

cross offline, Rust implementation against the Java specification: pair memo-ignore-args, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	2	violation	3.1	156	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-02
2	6	6	6	violation	3.4	158	10 min, 16 GB	3.4	0.1	0.0	in-memory	brain01, 2026-11-02
3	20	20	20	violation	3.6	158	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain01, 2026-11-02
4	70	70	70	violation	3.8	172	10 min, 16 GB	3.8	0.2	0.0	in-memory	brain01, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export

cross online, Java implementation (JMC) against the Rust specification export: pair **batched**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	0.6	164	1 h, 250 GB		brain05, 2026-11-02
2	6	4	conforms	0.6	170	1 h, 250 GB		brain05, 2026-11-02
3	20	8	conforms	0.7	148	1 h, 250 GB		brain05, 2026-11-02
4	70	16	conforms	0.7	168	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **delegating**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	0.6	150	1 h, 250 GB		brain05, 2026-11-02
2	6	6	conforms	0.7	148	1 h, 250 GB		brain05, 2026-11-02
3	20	20	conforms	0.8	152	1 h, 250 GB		brain05, 2026-11-02
4	70	70	conforms	1.6	288	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **memo**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	0.6	146	1 h, 250 GB		brain05, 2026-11-02
2	6	6	conforms	0.6	148	1 h, 250 GB		brain05, 2026-11-02
3	20	20	conforms	0.7	151	1 h, 250 GB		brain05, 2026-11-02
4	70	70	conforms	1.0	165	1 h, 250 GB		brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **batched-swap**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	2	2	2	violation	0.6	151	1 h, 250 GB	37	brain05, 2026-11-02
2	6	4	4	violation	0.6	156	1 h, 250 GB	41	brain05, 2026-11-02
3	20	8	8	violation	0.7	156	1 h, 250 GB	63	brain05, 2026-11-02
4	70	16	16	violation	0.8	167	1 h, 250 GB	53	brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **delegating-prev**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	2	2	2	violation	0.6	171	1 h, 250 GB	44	brain05, 2026-11-02
2	6	6	6	violation	0.7	150	1 h, 250 GB	63	brain05, 2026-11-02
3	20	20	20	violation	0.8	151	1 h, 250 GB	85	brain05, 2026-11-02
4	70	70	70	violation	1.9	283	1 h, 250 GB	127	brain05, 2026-11-02

cross online, Java implementation (JMC) against the Rust specification export: pair **memo-ignore-args**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	2	2	2	violation	0.6	146	1 h, 250 GB	41	brain05, 2026-11-02

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
2	6	6	6	violation	0.6	150	1 h, 250 GB	48	brain05, 2026-11-02
3	20	20	20	violation	0.7	166	1 h, 250 GB	56	brain05, 2026-11-02
4	70	70	70	violation	1.0	169	1 h, 250 GB	59	brain05, 2026-11-02

Not run (cross online, Java implementation (JMC) against the Rust specification export). planned but never scheduled (grid points beyond the ladder): 1 bag delegating, 1 bag memo-ignore-args, 1 bag batched-swap, 1 bag delegating-prev, 1 bag batched, 1 bag memo, 2 bag memo-ignore-args, 2 bag batched, 2 bag delegating-prev, 2 bag batched-swap, 2 bag memo, 2 bag delegating, 3 bag batched, 3 bag batched-swap, 3 bag memo, 3 bag delegating, 3 bag memo-ignore-args, 3 bag delegating-prev, 4 bag memo, 4 bag memo-ignore-args, 4 bag delegating, 4 bag batched, 4 bag batched-swap, 4 bag delegating-prev.

cross online, Rust implementation (TraceForge) against the Java specification export

cross online, Rust implementation (TraceForge) against the Java specification export: pair **batched**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
2	6	4	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
3	20	8	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
4	70	16	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **delegating**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
2	6	6	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
3	20	20	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
4	70	70	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **memo**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
2	6	6	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
3	20	20	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02
4	70	70	conforms	0.0	4	1 h, 250 GB		brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **batched-swap**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	2	2	2	violation	0.0	4	1 h, 250 GB	1	brain01, 2026-11-02
2	6	4	4	violation	0.0	4	1 h, 250 GB	1	brain01, 2026-11-02
3	20	8	8	violation	0.0	4	1 h, 250 GB	2	brain01, 2026-11-02
4	70	16	16	violation	0.0	4	1 h, 250 GB	1	brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair **delegating-prev**, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	2	2	2	violation	0.0	4	1 h, 250 GB	2	brain01, 2026-11-02
2	6	6	6	violation	0.0	4	1 h, 250 GB	2	brain01, 2026-11-02
3	20	20	20	violation	0.0	4	1 h, 250 GB	2	brain01, 2026-11-02
4	70	70	70	violation	0.0	4	1 h, 250 GB	2	brain01, 2026-11-02

cross online, Rust implementation (TraceForge) against the Java specification export: pair memo-ignore-args, model bag, expected violation

size	G_{spec}	G_{impl}	G_x	verdict	T	M	tier	first miss ms	run
1	2	2	2	violation	0.0	4	1 h, 250 GB	1	brain01, 2026-11-02
2	6	6	6	violation	0.0	4	1 h, 250 GB	2	brain01, 2026-11-02
3	20	20	20	violation	0.0	4	1 h, 250 GB	1	brain01, 2026-11-02
4	70	70	70	violation	0.0	4	1 h, 250 GB	2	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin

twins offline, Rust implementation against its Java twin: pair batched, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.1	161	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
2	4	4	conforms	3.4	173	10 min, 16 GB	3.3	0.0	0.0	in-memory	brain01, 2026-11-02
3	8	8	conforms	3.4	163	10 min, 16 GB	3.4	0.1	0.0	in-memory	brain01, 2026-11-02
4	16	16	conforms	3.6	166	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair batched-swap, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.2	171	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain01, 2026-11-02
2	4	4	conforms	3.2	159	10 min, 16 GB	3.2	0.1	0.0	in-memory	brain01, 2026-11-02
3	8	8	conforms	3.4	180	10 min, 16 GB	3.4	0.1	0.0	in-memory	brain01, 2026-11-02
4	16	16	conforms	3.6	164	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair delegating, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.2	158	10 min, 16 GB	3.1	0.0	0.1	in-memory	brain01, 2026-11-02
2	6	6	conforms	3.5	152	10 min, 16 GB	3.4	0.1	0.1	in-memory	brain01, 2026-11-02
3	20	20	conforms	3.8	172	10 min, 16 GB	3.6	0.1	0.2	in-memory	brain01, 2026-11-02
4	70	70	conforms	4.7	190	10 min, 16 GB	4.5	0.1	0.2	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair delegating-prev, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.3	156	10 min, 16 GB	3.3	0.1	0.0	in-memory	brain01, 2026-11-02
2	6	6	conforms	3.4	162	10 min, 16 GB	3.4	0.0	0.0	in-memory	brain01, 2026-11-02
3	20	20	conforms	3.8	170	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain01, 2026-11-02
4	70	70	conforms	4.7	231	10 min, 16 GB	4.7	0.1	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair **memo**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	2.9	160	10 min, 16 GB	2.9	0.1	0.0	in-memory	brain01, 2026-11-02
2	6	6	conforms	3.4	160	10 min, 16 GB	3.4	0.1	0.0	in-memory	brain01, 2026-11-02
3	20	20	conforms	3.6	164	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain01, 2026-11-02
4	70	70	conforms	3.8	158	10 min, 16 GB	3.8	0.1	0.0	in-memory	brain01, 2026-11-02

twins offline, Rust implementation against its Java twin: pair **memo-ignore-args**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.1	172	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain01, 2026-11-02
2	6	6	conforms	3.2	160	10 min, 16 GB	3.2	0.1	0.0	in-memory	brain01, 2026-11-02
3	20	20	conforms	3.6	168	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain01, 2026-11-02
4	70	70	conforms	4.2	166	10 min, 16 GB	3.8	0.2	0.4	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin

twins offline, Java implementation against its Rust twin: pair **batched**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.1	161	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02
2	4	4	conforms	3.3	173	10 min, 16 GB	0.0	3.3	0.0	in-memory	brain01, 2026-11-02
3	8	8	conforms	3.4	163	10 min, 16 GB	0.1	3.4	0.0	in-memory	brain01, 2026-11-02
4	16	16	conforms	3.6	166	10 min, 16 GB	0.1	3.6	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **batched-swap**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.1	171	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain01, 2026-11-02
2	4	4	conforms	3.2	159	10 min, 16 GB	0.1	3.2	0.0	in-memory	brain01, 2026-11-02
3	8	8	conforms	3.4	180	10 min, 16 GB	0.1	3.4	0.0	in-memory	brain01, 2026-11-02
4	16	16	conforms	3.6	164	10 min, 16 GB	0.1	3.6	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **delegating**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.1	158	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-02
2	6	6	conforms	3.4	152	10 min, 16 GB	0.1	3.4	0.0	in-memory	brain01, 2026-11-02
3	20	20	conforms	3.6	172	10 min, 16 GB	0.1	3.6	0.0	in-memory	brain01, 2026-11-02
4	70	70	conforms	4.5	190	10 min, 16 GB	0.1	4.5	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **delegating-prev**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.3	156	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain01, 2026-11-02
2	6	6	conforms	3.4	162	10 min, 16 GB	0.0	3.4	0.0	in-memory	brain01, 2026-11-02

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3	20	20	conforms	3.8	170	10 min, 16 GB	0.1	3.8	0.0	in-memory	brain01, 2026-11-02
4	70	70	conforms	4.7	231	10 min, 16 GB	0.1	4.7	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **memo**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	2.9	160	10 min, 16 GB	0.1	2.9	0.0	in-memory	brain01, 2026-11-02
2	6	6	conforms	3.4	160	10 min, 16 GB	0.1	3.4	0.0	in-memory	brain01, 2026-11-02
3	20	20	conforms	3.6	164	10 min, 16 GB	0.1	3.6	0.0	in-memory	brain01, 2026-11-02
4	70	70	conforms	3.8	158	10 min, 16 GB	0.1	3.8	0.0	in-memory	brain01, 2026-11-02

twins offline, Java implementation against its Rust twin: pair **memo-ignore-args**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.1	172	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain01, 2026-11-02
2	6	6	conforms	3.2	160	10 min, 16 GB	0.1	3.2	0.0	in-memory	brain01, 2026-11-02
3	20	20	conforms	3.6	168	10 min, 16 GB	0.1	3.6	0.0	in-memory	brain01, 2026-11-02
4	70	70	conforms	3.8	166	10 min, 16 GB	0.2	3.8	0.0	in-memory	brain01, 2026-11-02

context-free twins offline, Rust implementation against the context-free Java port

context-free twins offline, Rust implementation against the context-free Java port: pair **batched**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.1	153	10 min, 16 GB	3.1	0.1	0.0	in-memory	brain02, 2026-11-03
2	4	4	conforms	3.4	167	10 min, 16 GB	3.3	0.1	0.0	in-memory	brain02, 2026-11-03
3	8	8	conforms	3.8	158	10 min, 16 GB	3.4	0.1	0.4	in-memory	brain02, 2026-11-03
4	16	16	conforms	3.6	170	10 min, 16 GB	3.6	0.1	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **batched-swap**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.3	166	10 min, 16 GB	3.1	0.1	0.2	in-memory	brain02, 2026-11-03
2	4	4	conforms	2.9	168	10 min, 16 GB	2.9	0.1	0.0	in-memory	brain02, 2026-11-03
3	8	8	conforms	3.3	169	10 min, 16 GB	3.3	0.0	0.0	in-memory	brain02, 2026-11-03
4	16	16	conforms	3.5	161	10 min, 16 GB	3.5	0.2	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **delegating**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.0	162	10 min, 16 GB	2.9	0.1	0.0	in-memory	brain02, 2026-11-03
2	6	6	conforms	3.5	154	10 min, 16 GB	3.4	0.1	0.1	in-memory	brain02, 2026-11-03
3	20	20	conforms	3.6	162	10 min, 16 GB	3.6	0.0	0.0	in-memory	brain02, 2026-11-03
4	70	70	conforms	4.5	185	10 min, 16 GB	4.5	0.2	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **delegating-prev**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.1	171	10 min, 16 GB	3.1	0.0	0.0	in-memory	brain02, 2026-11-03
2	6	6	conforms	3.5	159	10 min, 16 GB	3.4	0.0	0.1	in-memory	brain02, 2026-11-03
3	20	20	conforms	3.6	174	10 min, 16 GB	3.6	0.0	0.0	in-memory	brain02, 2026-11-03
4	70	70	conforms	4.4	240	10 min, 16 GB	4.4	0.1	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **memo**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.4	170	10 min, 16 GB	3.3	0.1	0.1	in-memory	brain02, 2026-11-03
2	6	6	conforms	3.5	156	10 min, 16 GB	3.4	0.1	0.2	in-memory	brain02, 2026-11-03
3	20	20	conforms	3.8	169	10 min, 16 GB	3.6	0.0	0.2	in-memory	brain02, 2026-11-03
4	70	70	conforms	4.0	169	10 min, 16 GB	3.8	0.1	0.2	in-memory	brain02, 2026-11-03

context-free twins offline, Rust implementation against the context-free Java port: pair **memo-ignore-args**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.0	160	10 min, 16 GB	2.9	0.0	0.0	in-memory	brain02, 2026-11-03
2	6	6	conforms	3.4	154	10 min, 16 GB	3.4	0.1	0.1	in-memory	brain02, 2026-11-03
3	20	20	conforms	3.6	152	10 min, 16 GB	3.6	0.1	0.1	in-memory	brain02, 2026-11-03
4	70	70	conforms	4.1	164	10 min, 16 GB	3.8	0.0	0.3	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation

context-free twins offline, Java port against the Rust implementation: pair **batched**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.1	153	10 min, 16 GB	0.1	3.1	0.0	in-memory	brain02, 2026-11-03
2	4	4	conforms	3.3	167	10 min, 16 GB	0.1	3.3	0.0	in-memory	brain02, 2026-11-03
3	8	8	conforms	3.8	158	10 min, 16 GB	0.1	3.4	0.4	in-memory	brain02, 2026-11-03
4	16	16	conforms	3.6	170	10 min, 16 GB	0.1	3.6	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **batched-swap**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.3	166	10 min, 16 GB	0.1	3.1	0.2	in-memory	brain02, 2026-11-03
2	4	4	conforms	2.9	168	10 min, 16 GB	0.1	2.9	0.0	in-memory	brain02, 2026-11-03
3	8	8	conforms	3.3	169	10 min, 16 GB	0.0	3.3	0.0	in-memory	brain02, 2026-11-03
4	16	16	conforms	3.5	161	10 min, 16 GB	0.2	3.5	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **delegating**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	2.9	162	10 min, 16 GB	0.1	2.9	0.0	in-memory	brain02, 2026-11-03
2	6	6	conforms	3.4	154	10 min, 16 GB	0.1	3.4	0.1	in-memory	brain02, 2026-11-03

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
3	20	20	conforms	3.6	162	10 min, 16 GB	0.0	3.6	0.0	in-memory	brain02, 2026-11-03
4	70	70	conforms	4.5	185	10 min, 16 GB	0.2	4.5	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **delegating-prev**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.1	171	10 min, 16 GB	0.0	3.1	0.0	in-memory	brain02, 2026-11-03
2	6	6	conforms	3.5	159	10 min, 16 GB	0.0	3.4	0.1	in-memory	brain02, 2026-11-03
3	20	20	conforms	3.6	174	10 min, 16 GB	0.0	3.6	0.0	in-memory	brain02, 2026-11-03
4	70	70	conforms	4.4	240	10 min, 16 GB	0.1	4.4	0.0	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **memo**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.4	170	10 min, 16 GB	0.1	3.3	0.1	in-memory	brain02, 2026-11-03
2	6	6	conforms	3.5	156	10 min, 16 GB	0.1	3.4	0.2	in-memory	brain02, 2026-11-03
3	20	20	conforms	3.8	169	10 min, 16 GB	0.0	3.6	0.2	in-memory	brain02, 2026-11-03
4	70	70	conforms	4.0	169	10 min, 16 GB	0.1	3.8	0.2	in-memory	brain02, 2026-11-03

context-free twins offline, Java port against the Rust implementation: pair **memo-ignore-args**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	export spec	export impl	join	join kind	run
1	2	2	conforms	3.0	160	10 min, 16 GB	0.0	2.9	0.0	in-memory	brain02, 2026-11-03
2	6	6	conforms	3.4	154	10 min, 16 GB	0.1	3.4	0.1	in-memory	brain02, 2026-11-03
3	20	20	conforms	3.6	152	10 min, 16 GB	0.1	3.6	0.1	in-memory	brain02, 2026-11-03
4	70	70	conforms	4.1	164	10 min, 16 GB	0.0	3.8	0.3	in-memory	brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **batched**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	3.6	160	1 h, 250 GB		brain02, 2026-11-03
2	4	4	conforms	3.6	156	1 h, 250 GB		brain02, 2026-11-03
3	8	8	conforms	3.8	159	1 h, 250 GB		brain02, 2026-11-03
4	16	16	conforms	8.6	160	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair **batched-swap**, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	3.2	152	1 h, 250 GB		brain02, 2026-11-03
2	4	4	conforms	3.4	174	1 h, 250 GB		brain02, 2026-11-03
3	8	8	conforms	3.8	160	1 h, 250 GB		brain02, 2026-11-03
4	16	16	conforms	4.0	159	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair delegating, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	3.6	154	1 h, 250 GB		brain02, 2026-11-03
2	6	6	conforms	3.9	167	1 h, 250 GB		brain02, 2026-11-03
3	20	20	conforms	4.3	172	1 h, 250 GB		brain02, 2026-11-03
4	70	70	conforms	5.6	205	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair delegating-prev, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	3.3	160	1 h, 250 GB		brain02, 2026-11-03
2	6	6	conforms	4.1	168	1 h, 250 GB		brain02, 2026-11-03
3	20	20	conforms	4.5	166	1 h, 250 GB		brain02, 2026-11-03
4	70	70	conforms	5.8	235	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair memo, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	3.4	156	1 h, 250 GB		brain02, 2026-11-03
2	6	6	conforms	3.8	154	1 h, 250 GB		brain02, 2026-11-03
3	20	20	conforms	4.0	156	1 h, 250 GB		brain02, 2026-11-03
4	70	70	conforms	5.0	192	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Java port (JMC) against the Rust implementation's export: pair memo-ignore-args, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	3.4	155	1 h, 250 GB		brain02, 2026-11-03
2	6	6	conforms	3.6	170	1 h, 250 GB		brain02, 2026-11-03
3	20	20	conforms	4.0	162	1 h, 250 GB		brain02, 2026-11-03
4	70	70	conforms	4.8	192	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair batched, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	1.4	2	1 h, 250 GB		brain02, 2026-11-03
2	4	4	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3	8	8	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
4	16	16	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair batched-swap, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
2	4	4	conforms	0.0	2	1 h, 250 GB		brain02, 2026-11-03
3	8	8	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
4	16	16	conforms	0.0	2	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair delegating, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	0.0	2	1 h, 250 GB		brain02, 2026-11-03
2	6	6	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3	20	20	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
4	70	70	conforms	0.0	2	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair delegating-prev, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	0.0	2	1 h, 250 GB		brain02, 2026-11-03
2	6	6	conforms	0.0	2	1 h, 250 GB		brain02, 2026-11-03
3	20	20	conforms	0.0	2	1 h, 250 GB		brain02, 2026-11-03
4	70	70	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair memo, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
2	6	6	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
3	20	20	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
4	70	70	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03

context-free twins online, Rust implementation (TraceForge) against the Java port's export: pair memo-ignore-args, model bag, expected conforms

size	G_{spec}	G_{impl}	verdict	T	M	tier	first miss ms	run
1	2	2	conforms	0.0	2	1 h, 250 GB		brain02, 2026-11-03
2	6	6	conforms	0.0	2	1 h, 250 GB		brain02, 2026-11-03
3	20	20	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03
4	70	70	conforms	0.0	4	1 h, 250 GB		brain02, 2026-11-03